

Project 02 – Fine-Tuning Using Different Architectures

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****Framework:**** Hugging Face Transformers + TensorFlow
****GPU Environment:**** Google Colab (T4 GPU)

🧠 Task 1: Encoder Model (BERT) – Emotion Detection from Text
In this task, a ****BERT-Base (cased)**** encoder model was fine-tuned to perform multi-class emotion detection from text.

****Dataset:**** Kaggle – Emotion Categories (Neutral, Joy, Sadness, Anger)
Split: Train (17,640) | Validation (2,205) | Test (2,205)

****Preprocessing & Tokenization:****
- BERT tokenizer (`max\_length=128`, truncation, padding)
- Converted to TensorFlow datasets for GPU fine-tuning

****Training:****
- Optimizer: Adam (lr = 5e-5)
- Loss: Sparse Categorical Cross-Entropy
- Epochs: 3
- Accuracy: 58% → 75% (train), 65% (val)

****Evaluation:****
- Test Accuracy: 0.611
- Macro F1-score: 0.615
- Weighted F1-score: 0.617
- Test Loss: 0.974

****Observations:****
Best performance on **Joy** and **Anger** (F1 ≈ 0.75-0.77); mild confusion between **Sadness** and **Neutral**.

✅ Deliverables:
- Preprocessing & tokenization script
- Training & validation pipeline
- Evaluation metrics (Accuracy, F1, Confusion Matrix)
- Example predictions
****Status:**** ✅ Task 1 Complete

🍳 Task 2: Decoder Model (GPT-2) – Recipe Generation
****Objective:**** Fine-tune GPT-2 to generate complete cooking recipes from titles and ingredient lists.

****Dataset:**** [3A2M Extended Recipe Dataset (Kaggle)] (<https://www.kaggle.com/datasets/nazmussakibrupol/3a2mext/data>)

****Preprocessing:****

Merged `*title*`, `*ingredients*`, and `*directions*` into single text blocks.
Tokenizer: GPT-2 tokenizer (EOS token used as padding).

****Training:****

- Epochs: 3
- Learning rate: 5e-5
- Training loss $\approx$ 2.02
- Validation loss $\approx$ 1.96

****Evaluation:****

- ROUGE-1 = 0.24
- ROUGE-2 = 0.04
- ROUGE-L = 0.14
- BLEU = 0.013

****Observations:****

Generated recipes are coherent and ingredient-relevant. Some variation expected since recipe generation allows multiple correct outputs.

****Deliverables:****

- Tokenization & dataset formatting script
- Training loop for GPT-2
- Evaluation with ROUGE & BLEU
- Gradio web app for interactive recipe generation

****Status:****  Task 2 Complete

📖 Task 3: Encoder-Decoder Model (T5) – Text Summarization

****Objective:**** Fine-tune a T5 model for abstractive summarization of news articles from CNN/DailyMail dataset.

****Dataset Stats:**** Train = 3521 | Validation = 319 | Test = 287

****Model:**** T5-small (3 epochs, lr = 5e-5, batch size = 2)

****Evaluation:****

Metric	Score
ROUGE-1	10.63
ROUGE-2	1.83
ROUGE-L	9.48
Eval Loss	3.40

****Observations:****

Model learned concise summarization; small dataset limited performance. Best results achieved with beam search and controlled length penalty.

****Future Improvements:****

- Train longer or with T5-base
- Increase ``num_beams`` and apply ``length_penalty``
- Expand/clean dataset

****Status:****  Task 3 Complete

Summary

All three architectures were fine-tuned successfully:

Task	Model Type	Framework	Metric	Score
1	Encoder (BERT)	TensorFlow	F1	0.615
2	Decoder (GPT-2)	PyTorch	BLEU	0.013
3	Encoder-Decoder (T5)	TensorFlow	ROUGE-L	9.48

Each task demonstrates understanding of NLP architecture types and practical fine-tuning using the Hugging Face ecosystem.